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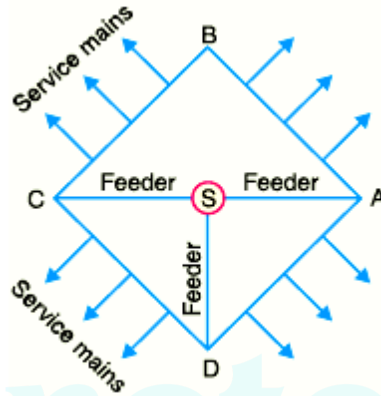
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MODULE 5

POWER DISTRIBUTION SYSTEMS

The electrical energy produced at the generating station is conveyed to the consumers through a network of transmission and distribution systems. It is often difficult to draw a line between the transmission and distribution systems of a large power system. In general, distribution system is that part of power system which distributes power to the consumers for utilization. That part of power system which distributes electric power for local use is known as distribution system. In general, the distribution system is the electrical system between the substation fed by the Transmission system and the consumer's meters. It generally consists of feeders, distributors, and service mains. Figure shows the single line diagram of a typical low tension distribution system.



(i) Feeders

A feeder is a conductor which connects the sub-station (or localized generating station) to the area where power is to be distributed. Generally, no tapings are taken from the feeder so that current in it remains the same throughout. The main consideration in the design of a feeder is the current carrying capacity.

(ii) Distributor

A distributor is a conductor from which tapings are taken for supply to the consumers. In the figure, AB, BC, CD and DA are the distributors. The current through a distributor is not constant because tapings are taken at various places along its length. While designing a distributor, voltage drop along its length is the main consideration since the statutory limit of voltage variations is $\pm 6\%$ of rated value at the consumers' terminals.

(iii) Service mains

A service mains is generally a small cable which connects the distributor to the consumers' terminals.

CLASSIFICATION OF DISTRIBUTION SYSTEMS

A distribution system may be classified according to;

(i) Nature of current

According to nature of current, distribution system may be classified as

- (a) d.c. Distribution system
- (b) a.c. Distribution system

Now-a-days, a.c. system is universally adopted for distribution of electric power as it is simpler and more economical than direct current method.

(ii) Type of construction

According to type of construction distribution system may be classified as

- (a) Overhead system
- (b) Underground system.

The overhead system is generally employed for distribution as it is 5 to 10 times cheaper than the equivalent underground system. In general, the underground system is used at places where overhead construction is impracticable or prohibited by the local laws

(iii) Scheme of connection

According to scheme of connection, the distribution system may be classified as

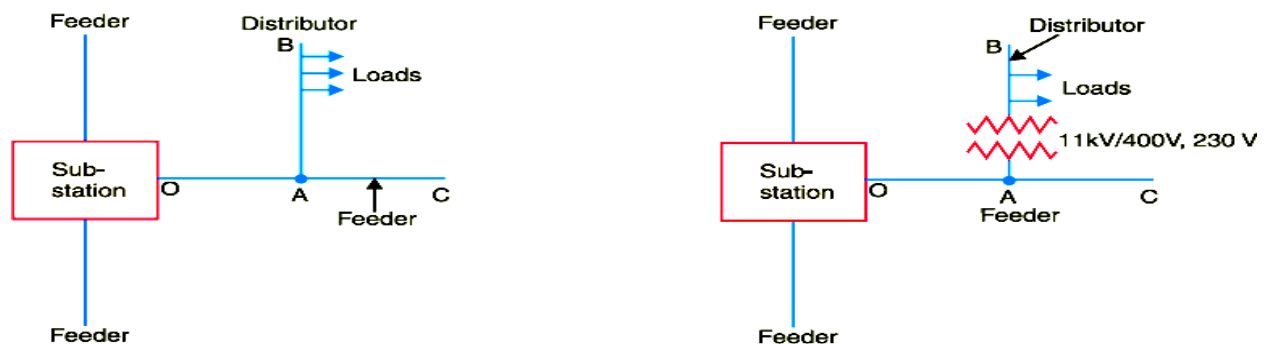
- (a) Radial system
- (b) Ring main system
- (c) Inter-connected system

CONNECTION SCHEMES OF DISTRIBUTION SYSTEM

All distribution of electrical energy is done by constant voltage system. In practice, the following distribution circuits are generally used:

(i) Radial System.

In this system, separate feeders radiate from a single substation and feed the distributors at one end only. Fig i). shows a single line diagram of a radial system for d.c. distribution where a feeder OC supplies a distributor A B at point A. Obviously, the distributor is fed at one end only i.e., point A is this case. Fig (ii) shows a single line diagram of radial system for a.c. distribution. The radial system is employed only when power is generated at low voltage and the substation is located at the centre of the load.



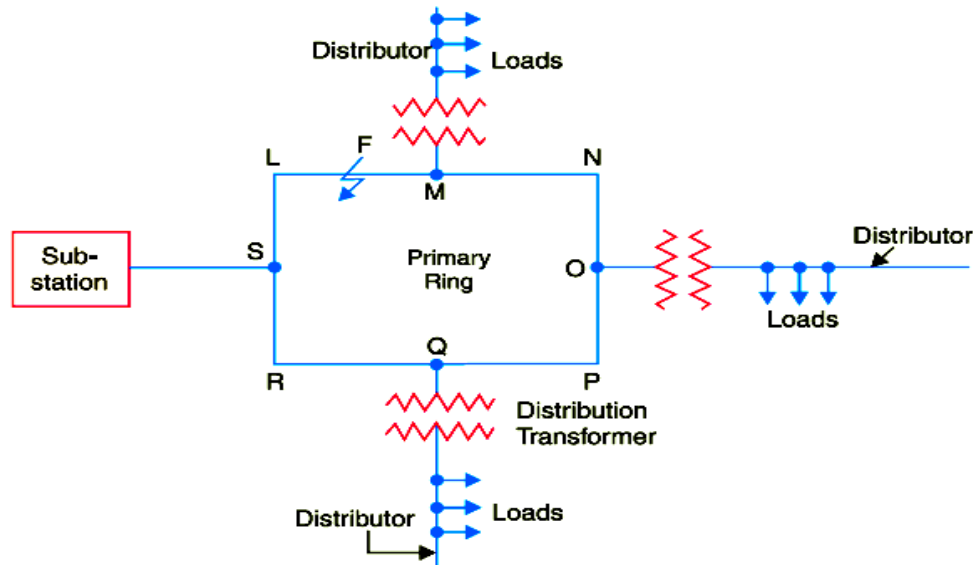
This is the simplest distribution circuit and has the lowest initial cost. However, it suffers from the following drawbacks:

- (a) The end of the distributor nearest to the feeding point will be heavily loaded.
- (b) The consumers are dependent on a single feeder and single distributor. Therefore, any fault on the feeder or distributor cuts off supply to the consumers who are on the side of the fault away from the substation.
- (c) The consumers at the distant end of the distributor would be subjected to serious voltage fluctuations when the load on the distributor changes.

Due to these limitations, this system is used for short distances only.

(ii) Ring main system.

In this system, the primaries of distribution transformers form a loop. The loop circuit starts from the substation bus-bars, makes a loop through the area to be served, and returns to the substation. Figure shows the single line diagram of ring main system for a.c. distribution where substation supplies to the closed feeder LMNOPQRS.

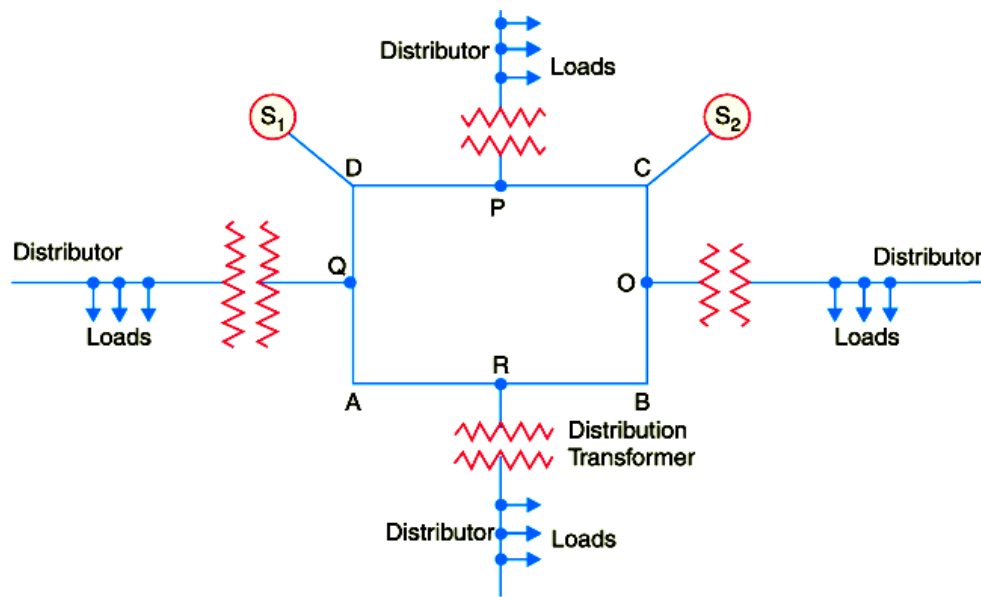


The distributors are tapped from different points M, O and Q of the feeder through distribution transformers. The ring main system has the following advantages:

- (a) There are less voltage fluctuations at consumer's terminals.
- (b) The system is very reliable as each distributor is fed via two feeders. In the event of fault on any section of the feeder, the continuity of supply is maintained. For example, suppose that fault occurs at any point F of section SLM of the feeder. Then section SLM of the feeder can be isolated for repairs and at the same time continuity of supply is maintained to all the consumers via the feeder SRQPONM.

(iii) Interconnected system.

When the feeder ring is energized by two or more than two generating stations or substations, it is called inter-connected system. Figure shows the single line diagram of interconnected system where the closed feeder ring ABCD is supplied by two substations S and S at points D and C respectively.



Distributors are connected to points O, P, Q and R of the feeder ring through distribution transformers. The interconnected system has the following advantages:

(a) It increases the service reliability.

(b) Any area fed from one generating station during peak load hours can be fed from the other generating station. This reduces reserve power capacity and increases efficiency of the system.

REQUIREMENTS OF A DISTRIBUTION SYSTEM

- (i) Proper voltage
- (ii) Availability of power on demand
- (iii) Reliability

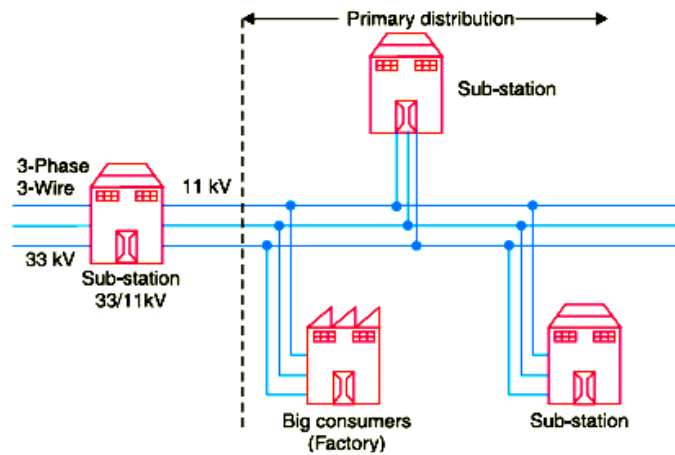
AC DISTRIBUTION

The a.c. distribution system is classified into

- (i) Primary distribution system and
- (ii) Secondary distribution system.

(i) Primary distribution system.

It is that part of a.c. distribution system which operates at voltages somewhat higher than general utilization and handles large blocks of electrical energy than the average low-voltage consumer uses. The voltage used for primary distribution depends upon the amount of power to be conveyed and the distance of the substation required to be fed. The most commonly used primary distribution voltages are 11 kV, 6.6 kV and 3.3 kV.



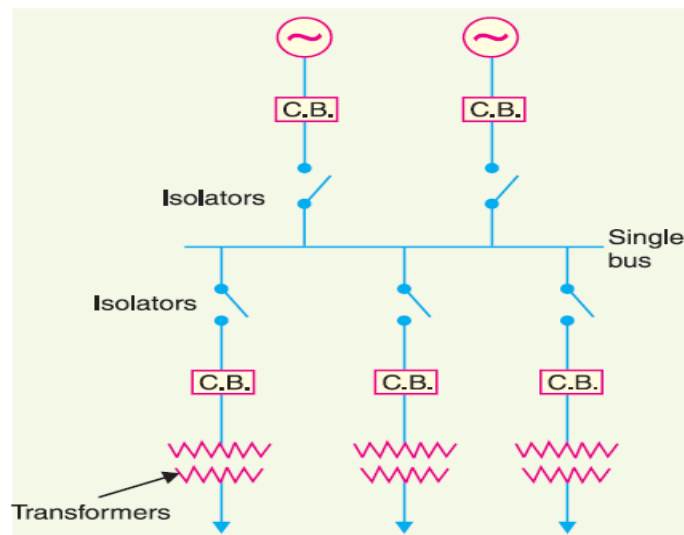
(ii) Secondary distribution system

The secondary distribution employs 415/240V, 3-phase, 4-wire system. The primary distribution circuit delivers power to various substations, called distribution sub-stations. The substations are situated near the consumers' localities and contain step-down transformers. At each distribution substation, the voltage is stepped down to 415V and power is delivered by 3-phase, 4-wire a.c. system. The voltage between any two phases is 415V and between any phase and neutral is 240V. The single phase domestic loads are connected between any one phase and the neutral, whereas 3-phase 415V motor loads are connected across 3-phase lines directly.

BUS BAR ARRANGEMENTS

When a number of generators or feeders operating at the same voltage have to be directly connected electrically, bus-bars are used as the common electrical component. Bus-bars are copper rods or thin walled tubes and operate at constant voltage.

(1) Single bus-bar system: The single bus bar system has the simplest design and is used for power stations. It is also used in small outdoor stations having relatively few outgoing or incoming feeders and lines. Fig shows the single bus-bar system for a typical power station. The generators, outgoing lines and transformers are connected to the bus-bar. Each generator and feeder is controlled by a circuit breaker. The isolators permit to isolate generators, feeders and circuit breakers from the bus-bar for maintenance. The chief advantages of this type of arrangement are low initial cost, less maintenance and simple operation.

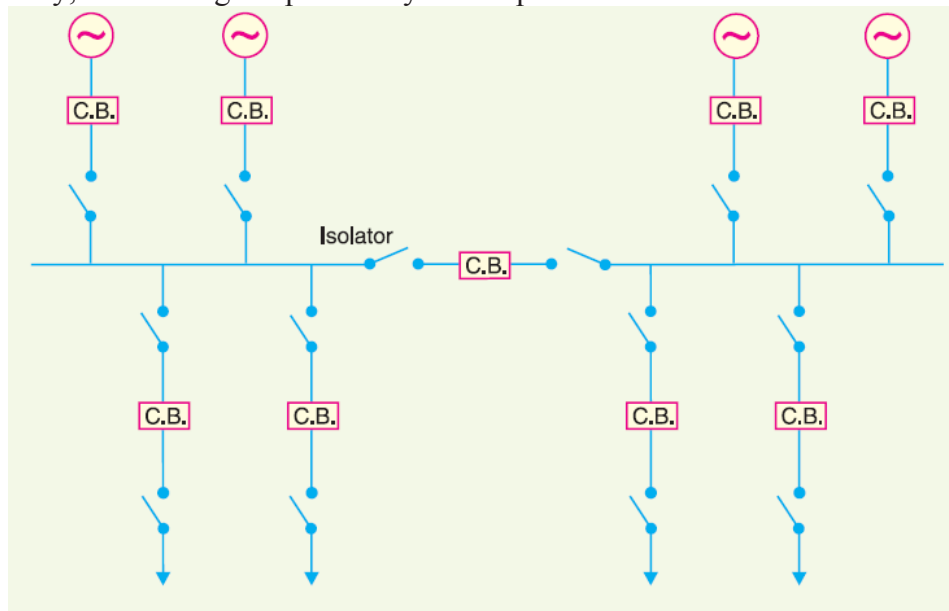


(2) Single bus-bar system with Sectionalisation. In large generating stations where several units are installed, it is a common practice to sectionalise the bus so that fault on any section of the bus-bar will not cause complete shutdown. Three principal advantages are claimed for this arrangement.

Firstly, if a fault occurs on any section of the bus-bar, that section can be isolated without affecting the supply to other sections.

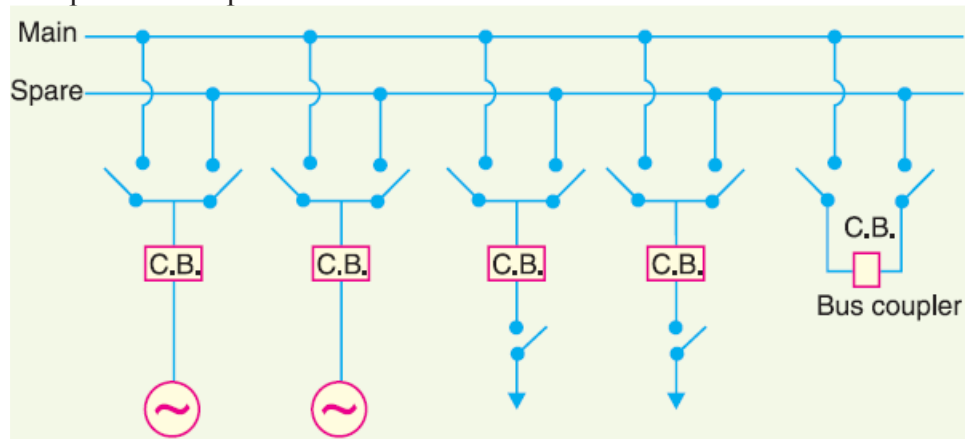
Secondly, if a fault occurs on any feeder, the fault current is much lower than with unsectionalised bus-bar. This permits the use of circuit breakers of lower capacity in the feeders.

Thirdly, repairs and maintenance of any section of the bus-bar can be carried out by de-energizing that section only, eliminating the possibility of complete shut-down.



(3) Duplicate bus-bar system. In large stations, it is important that breakdowns and maintenance should interfere as little as possible with continuity of supply. In order to achieve this objective, duplicate bus-bar system is used in important stations. Such a system consists of two bus-bars, a

“main bus-bar” and a “spare” bus-bar. Each generator and feeder may be connected to either bus-bar with the help of bus coupler which consists of a circuit breaker and isolators.

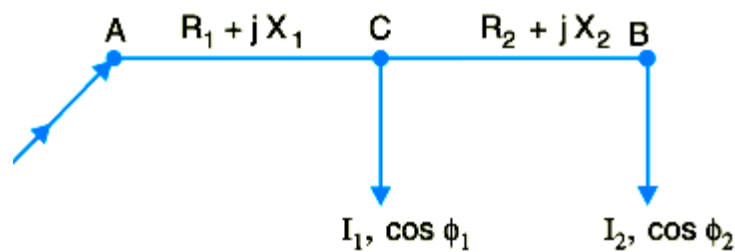


METHODS OF SOLVING A.C. DISTRIBUTION PROBLEMS

In a.c. distribution calculations, power factors of various load currents have to be considered since currents in different sections of the distributor will be the vector sum of load currents and not the arithmetic sum. The power factors of load currents may be given (i) w.r.t. receiving or sending end voltage or (ii) w.r.t. to load voltage itself. Each case shall be discussed separately.

(i) Power factors referred to receiving end voltage.

Consider an a.c. distributor AB with concentrated loads of I_1 and I_2 tapped off at points C and B as shown in Fig. Taking the receiving end voltage VB as the reference vector, let lagging power factors at C and B be $\cos \phi_1$ and $\cos \phi_2$ w.r.t. V_b . Let R_1, X_1 and R_2, X_2 be the resistance and reactance of sections AC and CB of the distributor.



Impedance of section AC , $\overrightarrow{Z_{AC}} = R_1 + j X_1$

Impedance of section CB , $\overrightarrow{Z_{CB}} = R_2 + j X_2$

Load current at point C , $\overrightarrow{I_1} = I_1 (\cos \phi_1 - j \sin \phi_1)$

Load current at point B , $\overrightarrow{I_2} = I_2 (\cos \phi_2 - j \sin \phi_2)$

Current in section CB , $\overrightarrow{I_{CB}} = \overrightarrow{I_2} = I_2 (\cos \phi_2 - j \sin \phi_2)$

Current in section AC , $\overrightarrow{I_{AC}} = \overrightarrow{I_1} + \overrightarrow{I_2}$
 $= I_1 (\cos \phi_1 - j \sin \phi_1) + I_2 (\cos \phi_2 - j \sin \phi_2)$

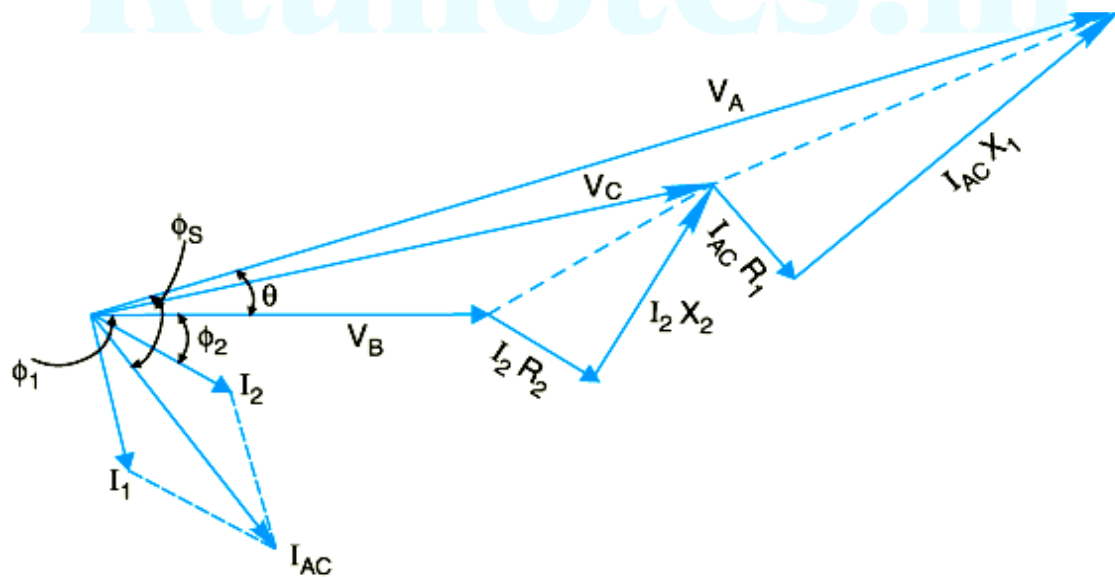
Voltage drop in section CB , $\overrightarrow{V_{CB}} = \overrightarrow{I_{CB}} \overrightarrow{Z_{CB}} = I_2 (\cos \phi_2 - j \sin \phi_2) (R_2 + j X_2)$

Voltage drop in section AC , $\overrightarrow{V_{AC}} = \overrightarrow{I_{AC}} \overrightarrow{Z_{AC}} = (\overrightarrow{I_1} + \overrightarrow{I_2}) \overrightarrow{Z_{AC}}$

$$= [I_1 (\cos \phi_1 - j \sin \phi_1) + I_2 (\cos \phi_2 - j \sin \phi_2)] [R_1 + j X_1]$$

Sending end voltage, $\overrightarrow{V_A} = \overrightarrow{V_B} + \overrightarrow{V_{CB}} + \overrightarrow{V_{AC}}$

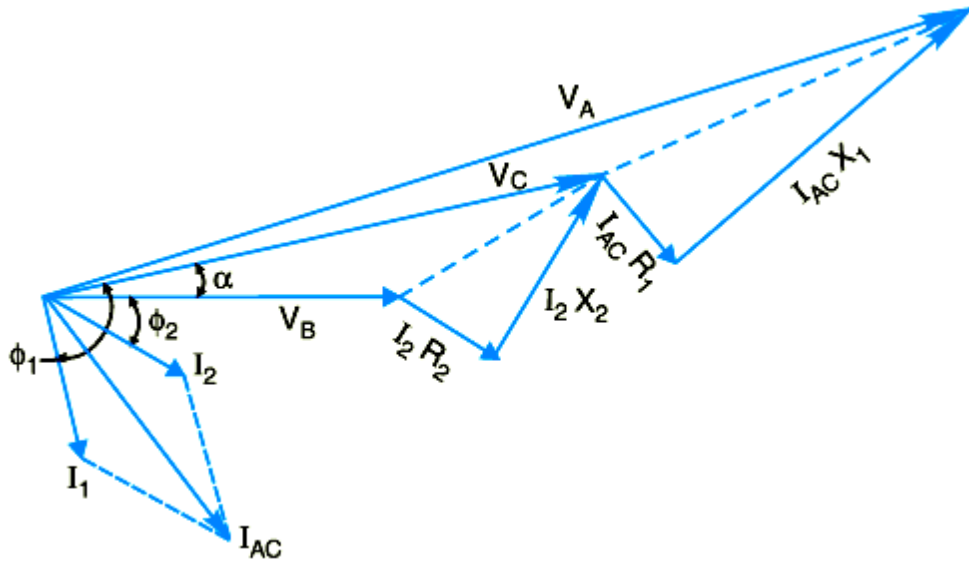
Sending end current, $\overrightarrow{I_A} = \overrightarrow{I_1} + \overrightarrow{I_2}$



The vector diagram of the a.c. distributor under these conditions is shown in Fig. Here, the receiving end voltage V_B is taken as the reference vector. As power factors of loads are given w.r.t. V_B , therefore, I_1 and I_2 lag behind V_B by ϕ_1 and ϕ_2 respectively.

(ii) Power factors referred to respective load voltages.

Suppose the power factors of loads in the previous Fig. are referred to their respective load voltages. Then ϕ_1 is the phase angle between V_C and I_1 and ϕ_2 is the phase angle between V_B and I_2 . The vector diagram under these conditions is shown in Fig



$$\text{Voltage drop in section } CB = \vec{I_2} \vec{Z_{CB}} = I_2 (\cos \phi_2 - j \sin \phi_2) (R_2 + j X_2)$$

$$\text{Voltage at point } C = \vec{V_B} + \text{Drop in section } CB = V_C \angle \alpha \text{ (say)}$$

$$\text{Now} \quad \vec{I_1} = I_1 \angle -\phi_1 \text{ w.r.t. voltage } V_C$$

$$\therefore \quad \vec{I_1} = I_1 \angle -(\phi_1 - \alpha) \text{ w.r.t. voltage } V_B$$

$$\text{i.e.} \quad \vec{I_1} = I_1 [\cos (\phi_1 - \alpha) - j \sin (\phi_1 - \alpha)]$$

$$\text{Now} \quad \vec{I_{AC}} = \vec{I_1} + \vec{I_2}$$

$$= I_1 [\cos (\phi_1 - \alpha) - j \sin (\phi_1 - \alpha)] + I_2 (\cos \phi_2 - j \sin \phi_2)$$

$$\text{Voltage drop in section } AC = \vec{I_{AC}} \vec{Z_{AC}}$$

$$\therefore \quad \text{Voltage at point } A = V_B + \text{Drop in } CB + \text{Drop in } AC$$

TYPES OF D.C. DISTRIBUTORS

The most general method of classifying d.c. distributors is the way they are fed by the feeders. On this basis, d.c. distributors are classified as:

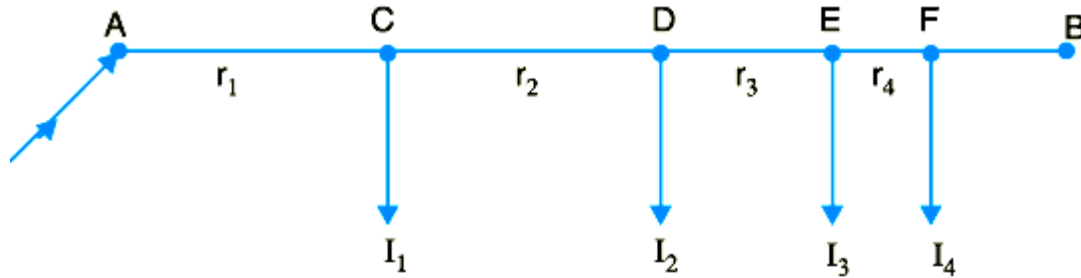
- (i) Distributor fed at one end
- (ii) Distributor fed at both ends
- (iii) Distributor fed at the centre

(iv) Ring distributor.

D.C. DISTRIBUTOR FED AT ONE END — CONCENTRATED LOADING

Fig. shows the single line diagram of a 2-wire d.c. distributor AB fed at one end A and having concentrated loads I_1, I_2, I_3 and I_4 tapped off at points C, D, E and F respectively.

Let r_1, r_2, r_3 and r_4 be the resistances of both wires (go and return) of the sections AC, CD, DE and EF of the distributor respectively.



$$\text{Current fed from point } A = I_1 + I_2 + I_3 + I_4$$

$$\text{Current in section } AC = I_1 + I_2 + I_3 + I_4$$

$$\text{Current in section } CD = I_2 + I_3 + I_4$$

$$\text{Current in section } DE = I_3 + I_4$$

$$\text{Current in section } EF = I_4$$

$$\text{Voltage drop in section } AC = r_1 (I_1 + I_2 + I_3 + I_4)$$

$$\text{Voltage drop in section } CD = r_2 (I_2 + I_3 + I_4)$$

$$\text{Voltage drop in section } DE = r_3 (I_3 + I_4)$$

$$\text{Voltage drop in section } EF = r_4 I_4$$

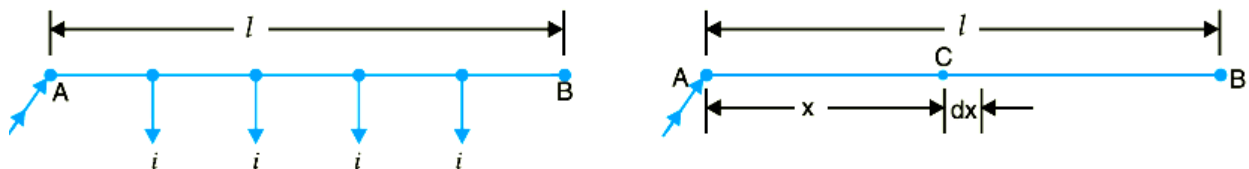
$\therefore$ Total voltage drop in the distributor

$$= r_1 (I_1 + I_2 + I_3 + I_4) + r_2 (I_2 + I_3 + I_4) + r_3 (I_3 + I_4) + r_4 I_4$$

It is easy to see that the minimum potential will occur at point F which is farthest from the feeding point A .

UNIFORMLY LOADED DISTRIBUTOR FED AT ONE END

Fig shows the single line diagram of a 2-wire d.c. distributor AB fed at one end A and loaded uniformly with i amperes per metre length. It means that at every 1 m length of the distributor, the load tapped is i amperes. Let l metres be the length of the distributor and r ohm be the resistance per metre run.



Consider a point C on the distributor at a distance x metres from the feeding point A as shown in Fig. Then current at point C is $= i l - i x$ amperes $= i(l - x)$ amperes

Now, consider a small length dx near point C . Its resistance is $r dx$ and the voltage drop over length dx is $d v = i(l - x) r dx = i r (l - x) dx$ Total voltage drop in the distributor upto point C is

$$V = \int_0^x i r (l - x) dx = i r \left(l x - \frac{x^2}{2} \right)$$

The voltage drop up to point B (i.e. over the whole distributor) can be obtained by putting $x = l$ in the above expression.

$\therefore$ Voltage drop over the distributor AB

$$\begin{aligned} &= i r \left(l \times l - \frac{l^2}{2} \right) \\ &= \frac{1}{2} i r l^2 = \frac{1}{2} (i l) (r l) \\ &= \frac{1}{2} I R \end{aligned}$$

where,

$i l = I$, the total current entering at point A

$r l = R$, the total resistance of the distributor

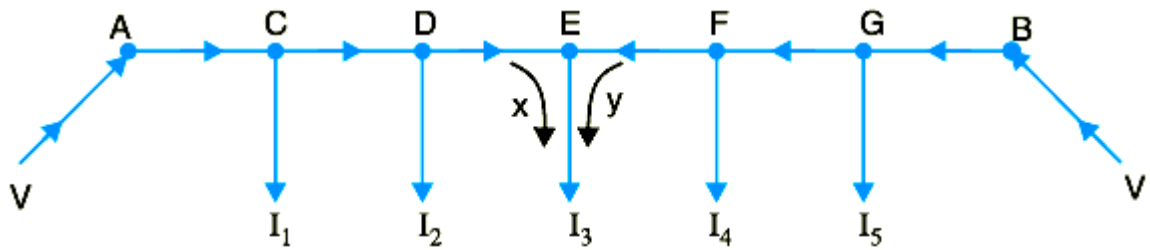
Thus, in a uniformly loaded distributor fed at one end, the total voltage drop is equal to that produced by the whole of the load assumed to be concentrated at the middle point.

DISTRIBUTOR FED AT BOTH ENDS — CONCENTRATED LOADING

Whenever possible, it is desirable that a long distributor should be fed at both ends instead of at one end only, since total voltage drop can be considerably reduced without increasing the cross-section of the conductor. The two ends of the distributor may be supplied with (i) equal voltages (ii) unequal voltages.

(i) Two ends fed with equal voltages.

Consider a distributor AB fed at both ends with equal voltages V volts and having concentrated loads I_1, I_2, I_3, I_4 and I_5 at points C, D, E, F and G respectively as shown in Fig. As we move away from one of the feeding points, say A , p.d. goes on decreasing till it reaches the minimum value at some load point, say E , and then again starts rising and becomes V volts as we reach the other feeding point B . All the currents tapped off between points A and E (minimum p.d. point) will be supplied from the feeding point A while those tapped off between B and E will be supplied from the feeding point B .



The current tapped off at point E itself will be partly supplied from A and partly from B. If these currents are x and y respectively, then,

$$I_3 = x + y$$

Therefore, we arrive at a very important conclusion that at the point of minimum potential, current comes from both ends of the distributor.

Point of minimum potential

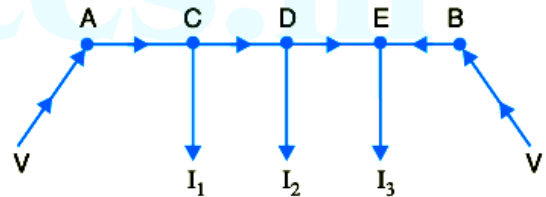
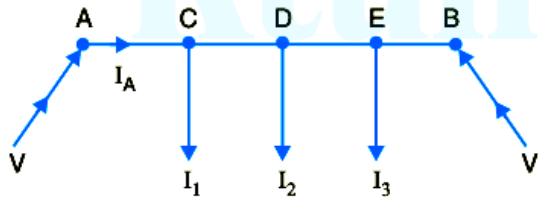
It is generally desired to locate the point of minimum potential. There is a simple method for it. Consider a distributor AB having three concentrated loads I_1 , I_2 and I_3 at points C , D and E respectively. Suppose that current supplied by feeding end A is I_A . Then current distribution in the various sections of the distributor can be worked out as shown in Fig.

$$I_{AC} = I_A;$$

$$I_{CD} = I_A - I_1$$

$$I_{DE} = I_A - I_1 - I_2;$$

$$I_{EB} = I_A - I_1 - I_2 - I_3$$



Voltage drop between A and B = Voltage drop over A B

$$\text{or } V - V = I_A R_{AC} + (I_A - I_1) R_{CD} + (I_A - I_1 - I_2) R_{DE} + (I_A - I_1 - I_2 - I_3) R_{EB}$$

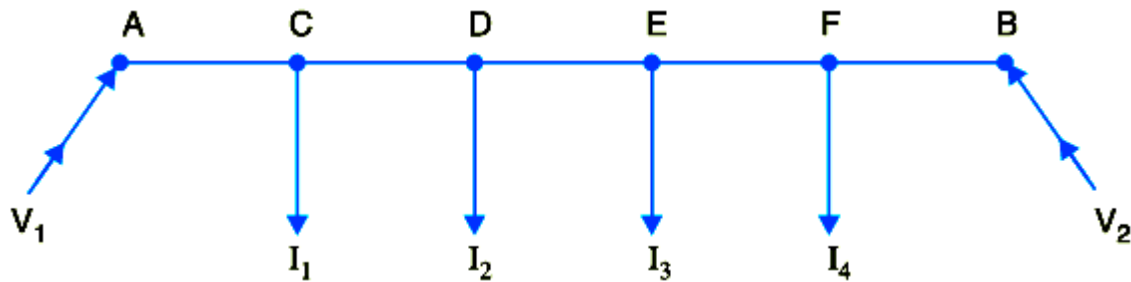
From this equation, the unknown I_A can be calculated as the values of other quantities are generally given. Suppose actual directions of currents in the various sections of the distributor are indicated as shown in Fig. The load point where the currents are coming from both sides of the distributor is the point of minimum potential i.e. point E in this case.

(ii) Two ends fed with unequal voltages.

Fig. shows the distributor AB fed with unequal voltages end A being fed at V_1 volts and end B at V_2 volts. The point of minimum potential can be found by following the same procedure as discussed above. Thus in this case,

Voltage drop between A and B = Voltage drop over A B

$$V_1 - V_2 = \text{Voltage drop over A B}$$



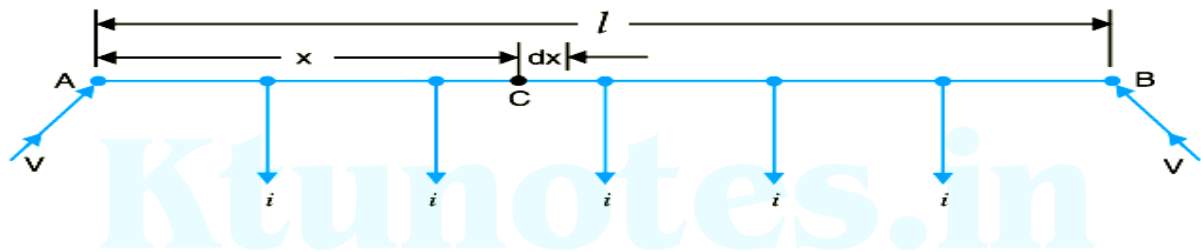
UNIFORMLY LOADED DISTRIBUTOR FED AT BOTH ENDS

There can be two cases viz. the distributor fed at both ends with (i) equal voltages (ii) unequal voltages. The two cases shall be discussed separately.

(i) Distributor fed at both ends with equal voltages.

Consider a distributor AB of length l metres, having resistance r ohms per metre run and with uniform loading of i amperes per metre run as shown in Fig. Let the distributor be fed at the feeding points A and B at equal voltages, say V volts. The total current supplied to the distributor is $i l$. As the two end voltages are equal, therefore, current supplied from each feeding point is $i l/2$ i.e.

Current supplied from each feeding point = $il/2$



Consider a point C at a distance x metres from the feeding point A . Then current at point C is $i l$

$$= \frac{i l}{2} - i x = i \left(\frac{l}{2} - x \right)$$

Now, consider a small length dx near point C . Its resistance is $r dx$ and the voltage drop over length dx is

$$dv = i \left(\frac{l}{2} - x \right) r dx = i r \left(\frac{l}{2} - x \right) dx$$

$$\begin{aligned} \therefore \text{Voltage drop upto point } C &= \int_0^x i r \left(\frac{l}{2} - x \right) dx = i r \left(\frac{l x}{2} - \frac{x^2}{2} \right) \\ &= \frac{i r}{2} (l x - x^2) \end{aligned}$$

Obviously, the point of minimum potential will be the mid-point. Therefore, maximum voltage drop will occur at mid-point i.e. where $x = l/2$.

$$\begin{aligned}
 \therefore \text{Max. voltage drop} &= \frac{i r}{2} (l x - x^2) \\
 &= \frac{i r}{2} \left(l \times \frac{l}{2} - \frac{l^2}{4} \right) \quad [\text{Putting } x = l/2] \\
 &= \frac{1}{8} i r l^2 = \frac{1}{8} (i l) (r l) = \frac{1}{8} I R
 \end{aligned}$$

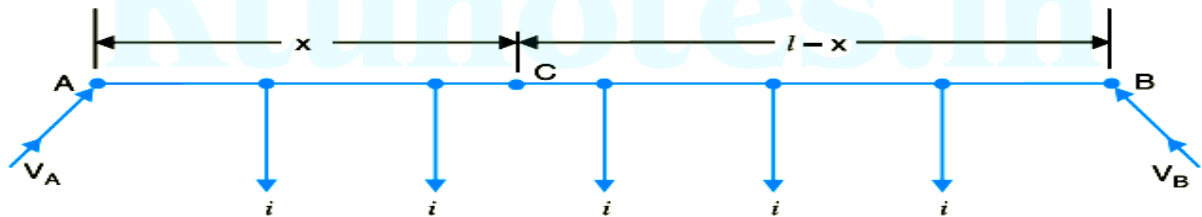
where $i l = I$, the total current fed to the distributor from both ends
 $r l = R$, the total resistance of the distributor

$$\text{Minimum voltage} = V - \frac{I R}{8} \text{ volts}$$

(ii) Distributor fed at both ends with unequal voltages.

Consider a distributor AB of length l metres having resistance r ohms per meter run and with a uniform loading of i amperes per metre run as shown in Fig. Let the distributor be fed from feeding points A and B at voltages V_A and V_B respectively. Suppose that the point of minimum potential C is situated at a distance x metres from the feeding point A. Then current supplied by the feeding point A will be $i x$.

$$\text{Voltage drop in section } AC = \frac{i r x^2}{2} \text{ volts}$$



As the distance of C from feeding point B is $(l - x)$, then, current fed from B is $i(l - x)$.

$$\begin{aligned}
 \therefore \text{Voltage drop in section } BC &= \frac{i r (l - x)^2}{2} \text{ volts} \\
 \text{Voltage at point C, } V_C &= V_A - \text{Drop over } AC \\
 &= V_A - \frac{i r x^2}{2} \quad \dots(i)
 \end{aligned}$$

$$\begin{aligned}
 \text{Also, voltage at point C, } V_C &= V_B - \text{Drop over } BC \\
 &= V_B - \frac{i r (l - x)^2}{2} \quad \dots(ii)
 \end{aligned}$$

From equations (i) and (ii), we get,

$$V_A - \frac{i r x^2}{2} = V_B - \frac{i r (l - x)^2}{2}$$

Solving the equation for x , we get,

$$x = \frac{V_A - V_B}{i r l} + \frac{l}{2}$$

As all the quantities on the right hand side of the equation are known, therefore, the point on the distributor where minimum potential occurs can be calculated.

AERIAL BUNDLED CABLES

Aerial bundled cables (also **aerial bundled conductors** or simply **ABC**) are overhead power lines using several insulated phase conductors bundled tightly together, usually with a bare neutral conductor. This contrasts with the traditional practice of using uninsulated conductors separated by air gaps. This variation of bundled conductors utilizes the same principles as overhead power lines, except that they are closer together to the point of touching but each conductor is surrounded by an insulating layer (except for the neutral line). The main objections to the traditional design are that the multiple conductors are considered unappealing, and external forces (such as high winds) can cause them to touch and short circuit.

Advantages

- Relative immunity to short circuits caused by external forces (wind, fallen branches), unless they abrade the insulation.
- Can stand in close proximity to trees/buildings and will not generate sparks if touched.
- Little to no tree trimming necessary
- Simpler installation, as crossbars and insulators are not required.
- More aesthetically appealing.
- Significantly improved safety for linespersons, particularly when working on live conductors.
- Electricity theft is made harder, and more obvious to detect.
- Less required maintenance and necessary inspections of lines.
- Improved reliability in comparison with both bare conductor overhead systems and underground systems. Insulated conductors prevent accidental contact and supply can be maintained temporarily in the event of a suspension system collapse.

Disadvantages

- Additional cost for the cable itself.
- Insulation degrades due to sun exposure, though the critical insulation between the wires is somewhat shielded from the sun.
- Shorter spans and more poles due to increased weight.
- Can lead to much longer repair times for installations in hilly areas due to much higher line weights requiring bigger and more specialized equipment to repair.

- Older installations are known to cause fires in areas where falling large trees or branches regularly cause breaks in lines and or in insulation leading to short circuits which can then lead to burning insulation dripping to ground and starting ground fires.

Grounding or Earthing

*The process of connecting the metallic frame (i.e. non-current carrying part) of electrical equipment or some electrical part of the system (e.g. neutral point in a star-connected system, one conductor of the secondary of a transformer etc.) to earth (i.e. soil) is called **grounding** or **earthing**.*

Grounding or earthing may be classified as :

(i) Equipment grounding **(ii)** System grounding.

Equipment grounding deals with earthing the non-current-carrying metal parts of the electrical equipment. On the other hand, system grounding means earthing some part of the electrical system e.g. earthing of neutral point of star-connected system in generating stations and sub-stations.

Equipment Grounding

*The process of connecting non-current-carrying metal parts (i.e. metallic enclosure) of the electrical equipment to earth (i.e. soil) in such a way that in case of insulation failure, the enclosure effectively remains at earth potential is called **equipment grounding**.*

System Grounding

*The process of connecting some electrical part of the power system (e.g. neutral point of a star connected system, one conductor of the secondary of a transformer etc.) to earth (i.e. soil) is called **system grounding**.*

Neutral Grounding

*The process of connecting neutral point of 3-phase system to earth (i.e. soil) either directly or through some circuit element (e.g. resistance, reactance etc.) is called **neutral grounding**.*

Neutral grounding provides protection to personal and equipment. It is because during earth fault, the current path is completed through the earthed neutral and the protective devices (e.g. a fuse etc.) operate to isolate the faulty conductor from the rest of the system.

POWER FACTOR

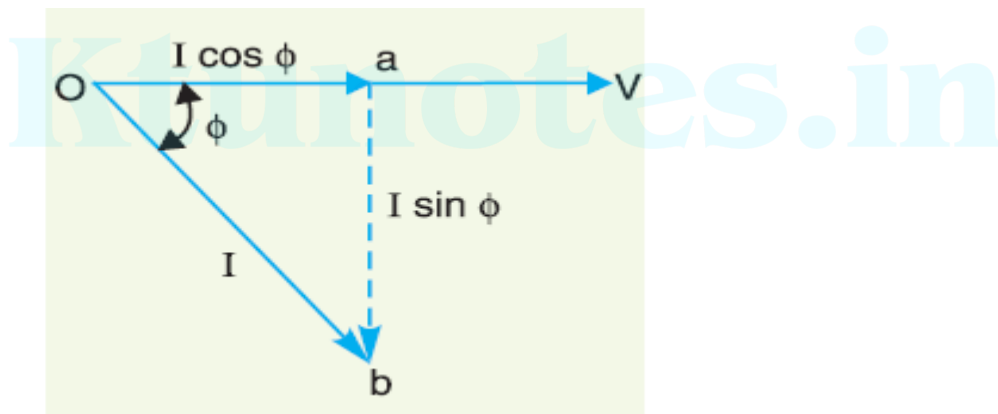
- The cosine of angle between voltage and current in an a.c. circuit is known as **power factor**.
- In an a.c. circuit, there is generally a phase difference ϕ between voltage and current. The term $\cos \phi$ is called the power factor of the circuit. If the circuit is inductive, the current lags behind the voltage and the power factor is referred to as lagging. However, in a capacitive circuit, current leads the voltage and power factor is said to be leading.

Consider an inductive circuit taking a lagging current I from supply voltage V ; the angle of lag being ϕ . The phasor diagram of the circuit is shown in Fig. The circuit current I can be resolved into perpendicular components, namely

(a) $I \cos \phi$ in phase with V

(b) $I \sin \phi$ 90 degree out of phase with V

The component $I \cos \phi$ is known as active or wattful component, whereas component $I \sin \phi$ is called the reactive or wattless component. The reactive component is a measure of the power factor. If the reactive component is small, the phase angle ϕ is small and hence power factor $\cos \phi$ will be high.



POWER TRIANGLE

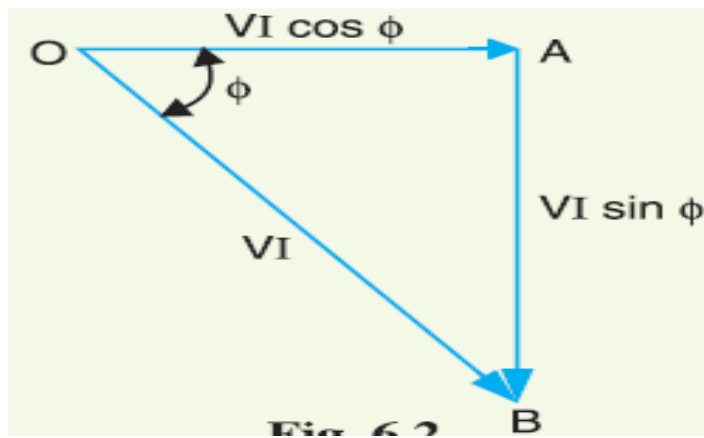
$OA = VI \cos \phi$ and represents the active power in watts or kW

$AB = VI \sin \phi$ and represents the reactive power in VAR or $kVAR$

$OB = VI$ and represents the apparent power in VA or kVA

$$OB^2 = OA^2 + AB^2 \text{ (apparent power)}^2 = \text{(active power)}^2 + \text{(reactive power)}^2$$

$$(\text{kVA})^2 = (\text{kW})^2 + (\text{kVAR})^2$$



$$\text{Power factor, } \cos \phi = \frac{OA}{OB} = \frac{\text{active power}}{\text{apparent power}} = \frac{\text{kW}}{\text{kVA}}$$

$$\text{kVAR} = \text{kVA} \sin \phi = \frac{\text{kW}}{\cos \phi} \sin \phi$$

$$\text{kVAR} = \text{kW} \tan \phi$$

DISADVANTAGES OF LOW PF

- The power factor plays an importance role in a.c. circuits since power consumed depends upon this factor.

$$P = V_L I_L \cos \phi \text{ (For single phase supply)}$$

$$I_L = \frac{P}{V_L \cos \phi}$$

$$P = \sqrt{3} V_L I_L \cos \phi \text{ (For 3 phase supply)}$$

$$I_L = \frac{P}{\sqrt{3} V_L \cos \phi}$$

- It is clear from above that for fixed power and voltage, the **load current is inversely proportional to the power factor**. Lower the power factor, higher is the load current and vice versa.

PF LESS THAN UNITY RESULTS IN FOLLOWING DISADVANTAGES

1. Large kVA rating of equipment

- The electrical machinery (e.g., alternators, transformers, switchgear) is always rated in kVA.
- It is clear that kVA rating of the equipment is inversely proportional to power factor. The smaller the power factor, the larger is the kVA rating. Therefore, at low power factor, the kVA rating of the equipment has to be made more, ***making the equipment larger and expensive.***

2. Greater conductor size

- To transmit or distribute a fixed amount of power at constant voltage, the conductor will have to carry ***more current*** at low power factor. This necessitates large conductor size.

3. Large copper losses-

- The large current at low power factor causes ***more I^2R losses*** in all the elements of the supply system. This results in poor efficiency.

4. Poor voltage regulation

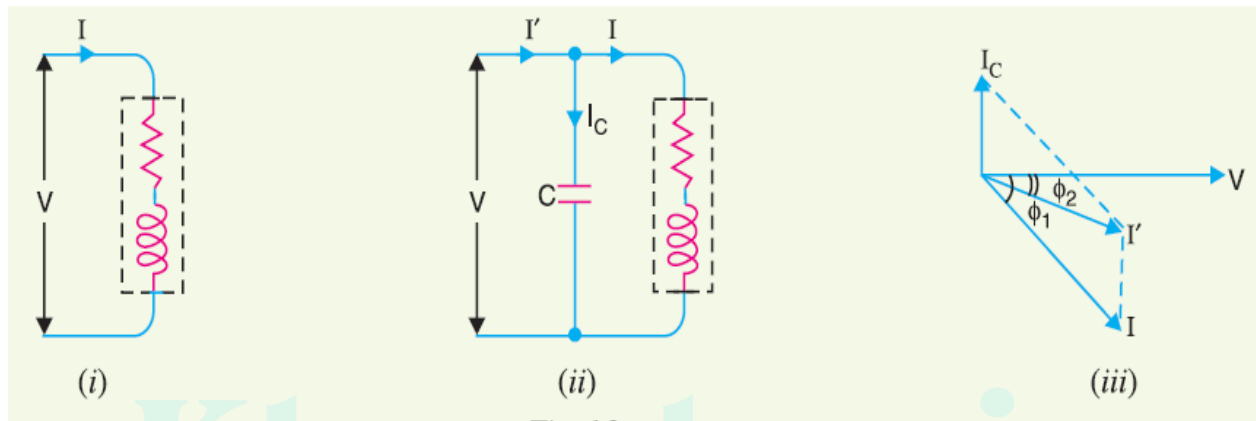
- The large current at low lagging power factor causes ***greater voltage drops*** in alternators, transformers, transmission lines and distributors. This results in the decreased voltage available at the supply end. Hence vol reg poor.

CAUSES OF LOW PF

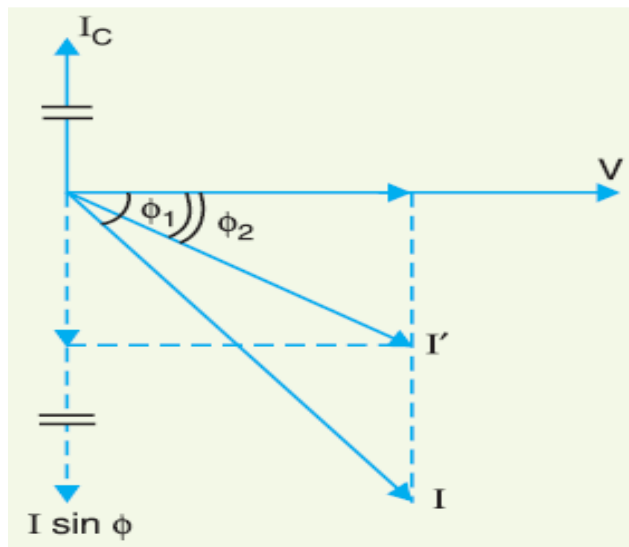
- (i) Most of the a.c. motors are of induction type (1ϕ and 3ϕ induction motors) which have low lagging power factor. These motors work at a power factor which is extremely small on light load (0.2 to 0.3) and rises to 0.8 or 0.9 at full load.
- (ii) Arc lamps, electric discharge lamps and industrial heating furnaces operate at low lagging power factor.
- (iii) The load on the power system is varying ; being high during morning and evening and low at other times. During low load period, supply voltage is increased which increases the magnetisation current. This results in the decreased power factor.

CALCULATION OF PF CORRECTION

The low power factor is mainly due to the fact that most of the power loads are inductive and, therefore, take lagging currents. In order to improve the power factor, some device taking leading power should be connected in parallel with the load. One of such devices can be a capacitor. The capacitor draws a leading current and partly or completely neutralises the lagging reactive component of load current. This raises the power factor of the load.



The capacitor C is connected in parallel with the load. The capacitor draws current I_c which leads the supply voltage by 90° . The resulting line current I' is the phasor sum of I and I_c and its angle of lag is ϕ_2 as shown in the phasor diagram. It is clear that ϕ_2 is less than ϕ_1 , so that $\cos \phi_2$ is greater than $\cos \phi_1$. Hence, the power factor of the load is improved.



From the phasor diagram, it is clear that after p.f. correction, the lagging reactive component of the load is reduced to $I' \sin \phi_2$.

Obviously, $I' \sin \phi_2 = I \sin \phi_1 - I_C$

or $I_C = I \sin \phi_1 - I' \sin \phi_2$

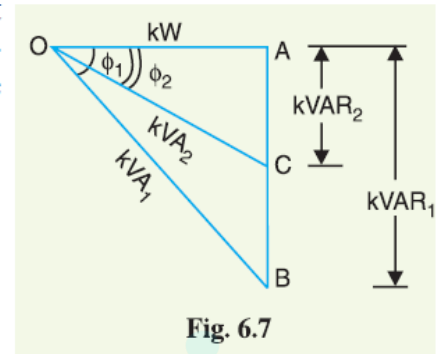
$\therefore$ Capacitance of capacitor to improve p.f. from $\cos \phi_1$ to $\cos \phi_2$

$$= \frac{I_C}{\omega V} \quad \left(\because X_C = \frac{V}{I_C} = \frac{1}{\omega C} \right)$$

Power triangle. The power factor correction can also be illustrated from power triangle. Thus referring to Fig. 6.7, the power triangle OAB is for the power factor $\cos \phi_1$, whereas power triangle OAC is for the improved power factor $\cos \phi_2$. It may be seen that active power (OA) does not change with power factor improvement. However, the lagging kVAR of the load is reduced by the p.f. correction equipment, thus improving the p.f. to $\cos \phi_2$.

Leading kVAR supplied by p.f. correction equipment

$$\begin{aligned} &= BC = AB - AC \\ &= \text{kVAR}_1 - \text{kVAR}_2 \\ &= OA (\tan \phi_1 - \tan \phi_2) \\ &= \text{kW} (\tan \phi_1 - \tan \phi_2) \end{aligned}$$



Knowing the leading kVAR supplied by the p.f. correction equipment, the desired results can be obtained.

PF IMPROVEMENT

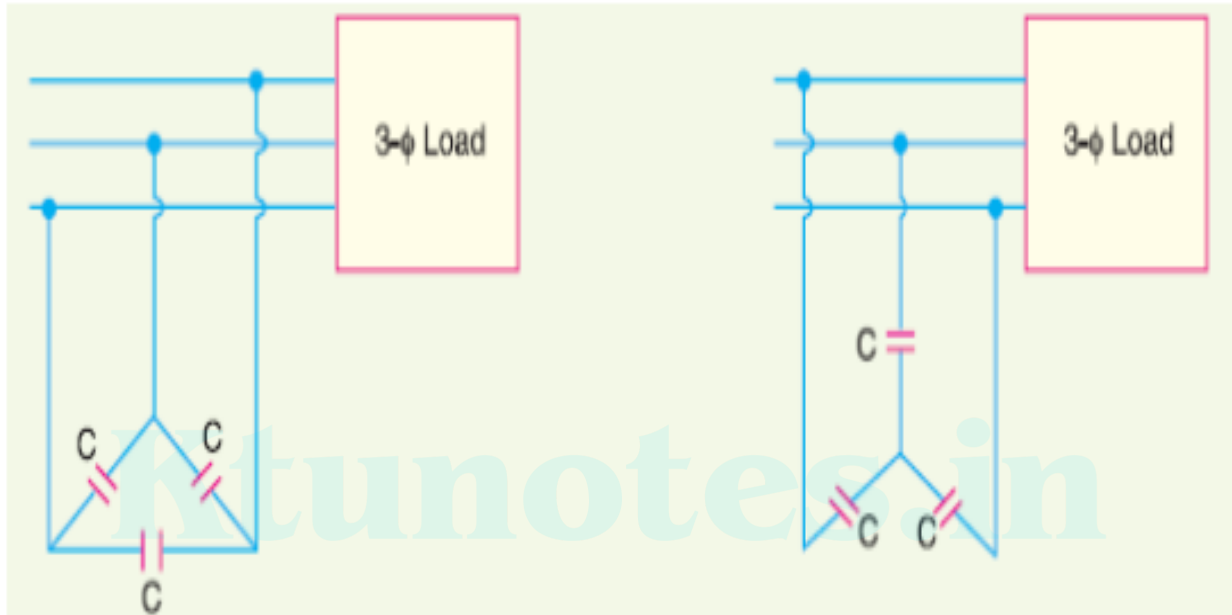
- The electrical energy is almost exclusively generated, transmitted and distributed in the form of alternating current. Therefore, the question of power factor immediately comes into picture. Most of the loads (*e.g.* induction motors, arc lamps) are inductive in nature and hence have low lagging power factor. The low power factor is highly undesirable as it causes an increase in current, resulting in additional losses of active power in all the elements of power system from power station generator down to the utilisation devices. In order to ensure most favourable conditions for a supply system from engineering and economical standpoint, it is important to have power factor as close to unity as possible.

PF IMPROVEMENT EQUIPMENT

- 1. Static capacitors.
- 2. Synchronous condenser.
- 3. Phase advancers.

Static capacitor.

- The power factor can be improved by connecting capacitors in parallel with the equipment operating at lagging power factor.
- The capacitor (generally known as static capacitor) draws a leading current and partly or completely neutralises the lagging reactive component of load current. This raises the power factor of the load.
- For three-phase loads, the capacitors can be connected in **delta or star** as shown in Fig.
- Static capacitors are invariably used for power factor improvement in factories.



Advantages

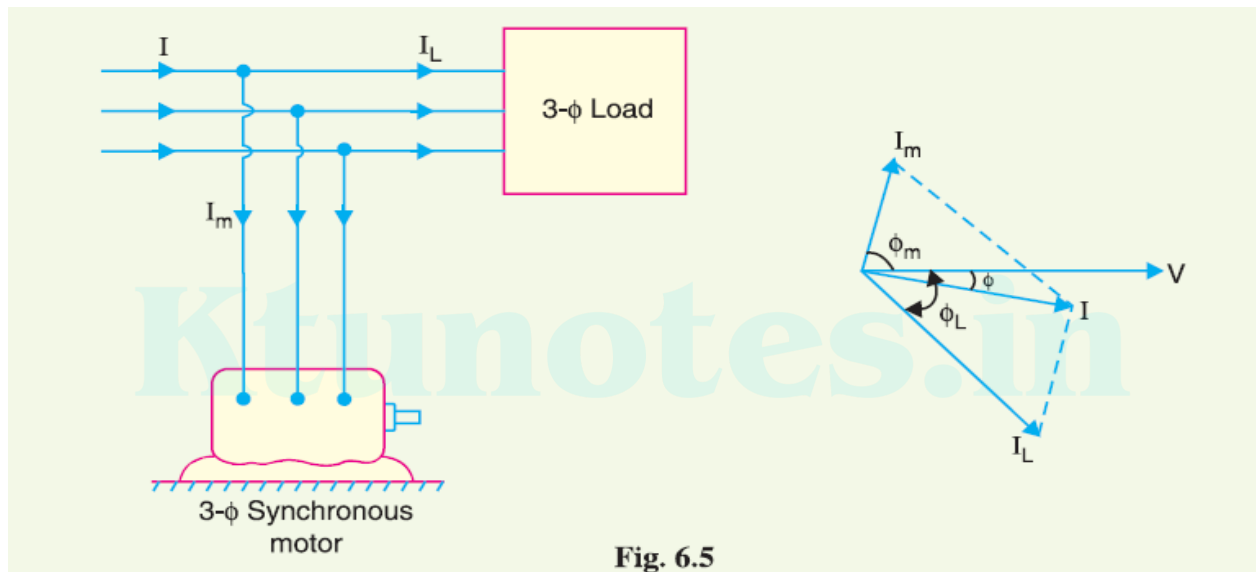
- (i) They have low losses.
- (ii) They require little maintenance as there are no rotating parts.
- (iii) They can be easily installed as they are light and require no foundation.
- (iv) They can work under ordinary atmospheric conditions.

Disadvantages

- (i) They have short service life ranging from 8 to 10 years.
- (ii) They are easily damaged if the voltage exceeds the rated value.
- (iii) Once the capacitors are damaged, their repair is uneconomical.

SYNCHRONOUS CONDENSER

- A synchronous motor takes a leading current when over-excited and, therefore, behaves as a capacitor. An over-excited synchronous motor running on no load is known as *synchronous condenser*.
- When such a machine is connected in parallel with the supply, it takes a leading current which partly neutralises the lagging reactive component of the load. Thus the power factor is improved.
- The 3 ϕ load takes current I_L at low lagging power factor $\cos \phi_L$. The synchronous condenser takes a current I_m which leads the voltage by an angle ϕ_m . The resultant current I is the phasor sum of I_m and I_L and lags behind the voltage by an angle ϕ .
- It is clear that ϕ is less than ϕ_L so that $\cos \phi$ is greater than $\cos \phi_L$.
- Thus the power factor is increased from $\cos \phi_L$ to $\cos \phi$.



Tariff

The rate at which electrical energy is supplied to a consumer is known as **tariff**.

Objectives of tariff. Like other commodities, electrical energy is also sold at such a rate so that it not only returns the cost but also earns reasonable profit. Therefore, a tariff should include the following items :

- (i) Recovery of cost of producing electrical energy at the power station.
- (ii) Recovery of cost on the capital investment in transmission and distribution systems.
- (iii) Recovery of cost of operation and maintenance of supply of electrical energy e.g., metering equipment, billing etc.
- (iv) A suitable profit on the capital investment.

Desirable Characteristics of a Tariff

A tariff must have the following desirable characteristics :

- (i) **Proper return** : The tariff should be such that it ensures the proper return from each consumer. The total receipts from the consumers must be equal to the cost of producing and supplying

electrical energy plus reasonable profit. This will enable the electric supply company to ensure continuous and reliable service to the consumers.

(ii) Fairness : The tariff must be fair so that different types of consumers are satisfied with the rate of charge of electrical energy.

(iii) Simplicity : The tariff should be simple so that an ordinary consumer can easily understand it. A complicated tariff may cause an opposition from the public which is generally distrustful of supply companies.

(iv) Reasonable profit : The profit element in the tariff should be reasonable. An electric supply company is a public utility company and generally enjoys the benefits of monopoly. Therefore, the investment is relatively safe due to non-competition in the market.

(v) Attractive : The tariff should be attractive so that a large number of consumers are encouraged to use electrical energy. Efforts should be made to fix the tariff in such a way so that consumers can pay easily.

Types of Tariff

1. Simple tariff. *When there is a fixed rate per unit of energy consumed, it is called a **simple tariff** or **uniform rate tariff**.*

In this type of tariff, the price charged per unit is constant *i.e.*, it does not vary with increase or decrease in number of units consumed. The consumption of electrical energy at the consumer's terminals is recorded by means of an energy meter. This is the simplest of all tariffs and is readily understood by the consumers.

2. Flat rate tariff. *When different types of consumers are charged at different uniform per unit rates, it is called a **flat rate tariff**.* In this type of tariff, the consumers are grouped into different classes and each class of consumers is charged at a different uniform rate.

3. Block rate tariff. *When a given block of energy is charged at a specified rate and the succeeding blocks of energy are charged at progressively reduced rates, it is called a **block rate tariff**.*

In block rate tariff, the energy consumption is divided into blocks and the price per unit is fixed in each block. The price per unit in the first block is the highest and it is progressively reduced for the succeeding blocks of energy.

4. Two-part tariff. *When the rate of electrical energy is charged on the basis of maximum demand of the consumer and the units consumed, it is called a **two-part tariff**.*

In two-part tariff, the total charge to be made from the consumer is split into two components *viz.*, fixed charges and running charges. The fixed charges depend upon the maximum demand of the consumer while the running charges depend upon the number of units consumed by the consumer.

$$\text{Total charges} = \text{Rs } (b \times \text{kW} + c \times \text{kWh})$$

$$b = \text{charge per kW of maximum demand}$$

$$c = \text{charge per kWh of energy consumed}$$

5. Maximum demand tariff. It is similar to two-part tariff with the only difference that the maximum demand is actually measured by installing maximum demand meter in the premises of the consumer. This removes the objection of two-part tariff where the maximum demand is assessed merely on the basis of the rateable value. This type of tariff is mostly applied to big consumers.

However, it is not suitable for a small consumer (e.g., residential consumer) as a separate maximum demand meter is required.

6. Power factor tariff. *The tariff in which power factor of the consumer's load is taken into consideration is known as **power factor tariff**.*

In an a.c. system, power factor plays an important role. A low power factor increases the rating of station equipment and line losses. Therefore, a consumer having low power factor must be penalised.

7. Three-part tariff. *When the total charge to be made from the consumer is split into three parts viz., fixed charge, semi-fixed charge and running charge, it is known as a **three-part tariff**.*

$$\text{Total charge} = \text{Rs } (a + b \times \text{kW} + c \times \text{kWh})$$

a = fixed charge made during each billing period. It includes interest and depreciation on the cost of secondary distribution and labour cost of collecting revenues,

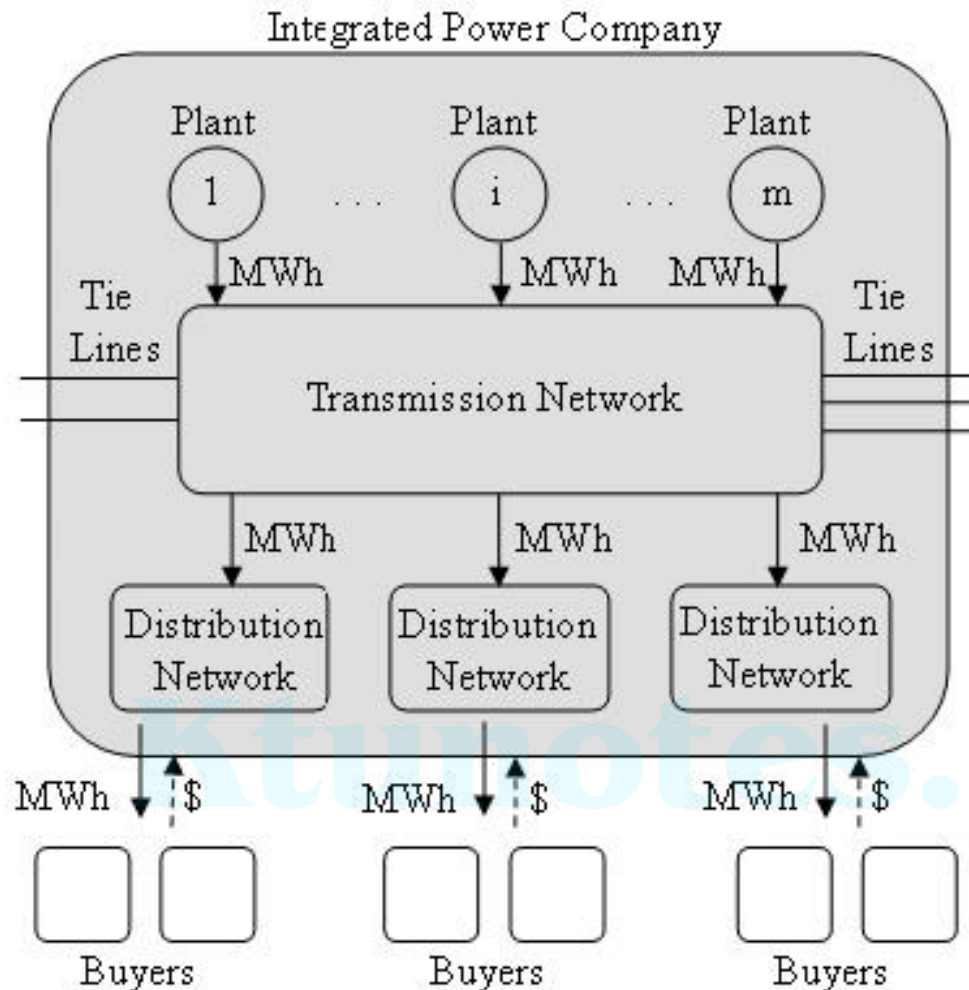
b = charge per kW of maximum demand,

c = charge per kWh of energy consumed.

ENERGY MARKETS

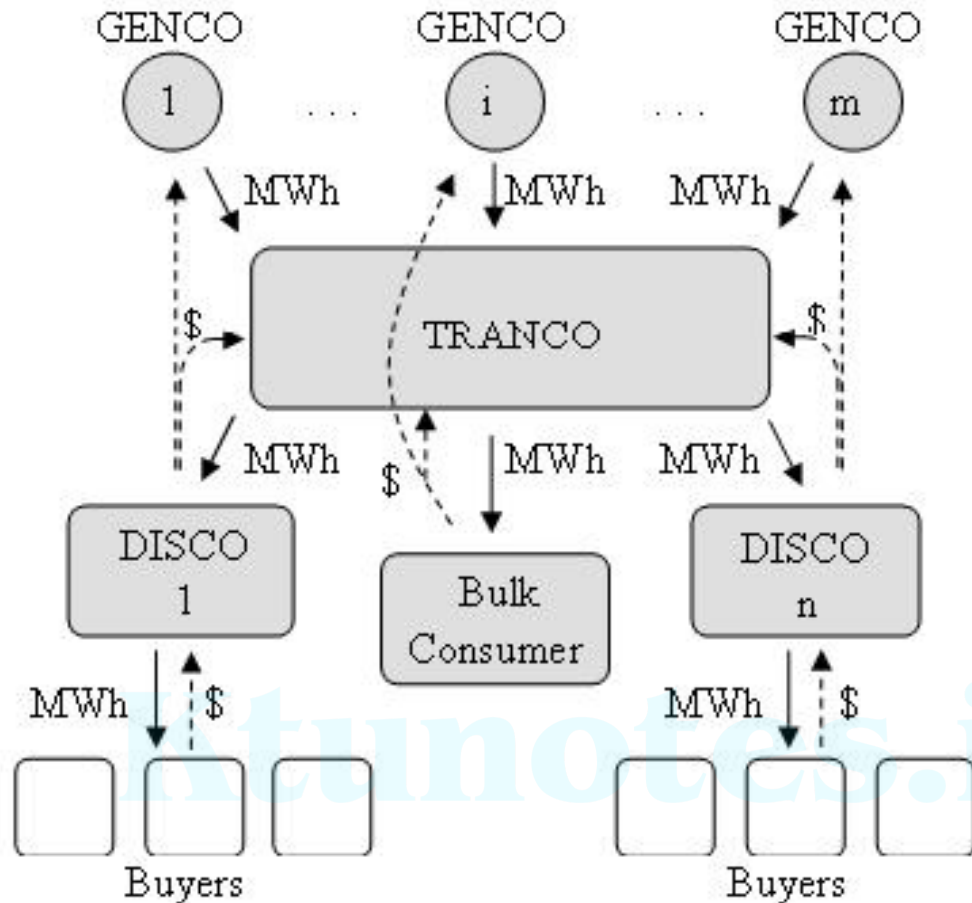
- Power systems historically evolved into structures with large centrally located power plants and highly meshed transmission networks
- It was natural that vertically integrated company would serve all the elements of the system
- Electric power industries monopolized the way in which they are operated and controlled till the late 1970s.
- Industries like aviation and telecommunication demonstrated that vertically integrated monopoly could not provide services as efficiently as the competitive firms
- Analogous to these industries, electric power industry also introduce competition by establishing a restructured environment
- Restructuring of electrical power system is the process of unbundling the services provided by state owned power industry, which was previously a vertically integrated monopoly
- Restructured electric power industry provides a market based system that allows
 - competition in energy trading,
 - open access of infrastructure and
 - improved reliability of service with high efficiency

VERTICALLY INTEGRATED OR REGULATED POWER INDUSTRY



- For many decades since their inception, electric power industries have monopolized the way in which they are operated controlled the main components of the system: generation, transmission and distribution, and sold electricity to customers within their geographical region
- This vertically integrated utility was obligated to sell electric power and meet the needs of its customers. These are publicly owned and not operated for profit, and the tariffs are set by the regulatory organizations
- Power grid has been formed by interconnecting the utilities through transmission network
- Tie-lines facilitate transfer of energy from one control area to another to share respective capacities, increase the system reliability and transfer power at the time of emergency or for economic reasons.

RESTRUCTURED OR DEREGULATED POWER INDUSTRY



- Recent reforms in energy sector have created a new environment of competitive market in which the market participants have open access to the system infrastructure.
- In restructured power system structure, generating companies (GENCOs) are owned by different power suppliers and they compete to sell energy to the consuming utilities at competitive prices.
- Transmission company (TRANSCOs) will transmit power to the load centers over high voltage lines. Distribution companies (DISCOs) will distribute power at the whole-sale or retail level to the customers.
- A competitive electricity market and open access of transmission infrastructure necessitated an independent control of the power grid
- Independent system operator (ISO) ensures fair and non-discretionary access to transmission services, maintains the real-time operation and reliability of power system.
- Power exchange (PX) is an independent non-profit entity, which basically provides a competitive marketplace by running electronic auction and thereby permitting different participants to sell/buy energy and other associated ancillary services in a competitive way based on quantity bids and prices.
- Through auction, the PX establishes a market clearing price (MCP) for each hour of a day for the trades between buyers and sellers.

- The PX clears the market only after balancing of loads and generation resources and ensuring the corresponding schedule for reliable transmission.

ENTITIES IN RESTRUCTURED ELECTRICITY MARKETS OR MARKET PARTICIPANTS

Generator companies (gencos)

- The gencos produce and sell electricity
- This may be individual generating units or more often to a group of generating units within a single company
- objective - producing power - independent power producers (IPP)

Transmission Companies (transcos)

- Entities, which own and operate the transmission network
- Prime responsibility is to transport the electricity from the generators to the customer, and making available the transmission system to all entities in the system
- For their services, they levy a transmission tariff

Distribution Companies (discos)

- The distribution companies are usually those entities owning and operating the local distribution network in an area
- They buy wholesale electricity either through the spot-markets or through direct contracts with gencos and supply electricity to the end-use customers

Customers (Consumers)

- A customer is an entity consuming electricity
- In deregulated markets, the customer has several options for buying electricity
- It may choose to buy electricity from the spot-market by bidding for purchase, or may buy directly from a genco or even from the local distribution company

Independent System Operator (ISO)

- Responsible for ensuring the reliability and security of the system
- It is an independent authority
- does not participate in the electricity market trades
- It usually does not own generating resources, except for some reserve capacity in certain cases
- ISO procures various services such as supply of emergency reserves, or reactive power from other entities in the system

Market Operator (Power Exchange)

- Responsible for the operation of the electricity market trading
- It receives bid offers from market participants
- Determines the market price
- Different trading schemes such as hourly trading for the next day or trading in futures - weeks, months or years ahead
- Marketplace to trade energy and other services in a competitive manner. Independent, non-government and non-profit entity
- Permits different participants to sell and buy energy and other services in a competitive way
- Px guarantees an equal and non-discriminatory access and competitive opportunity to all participants

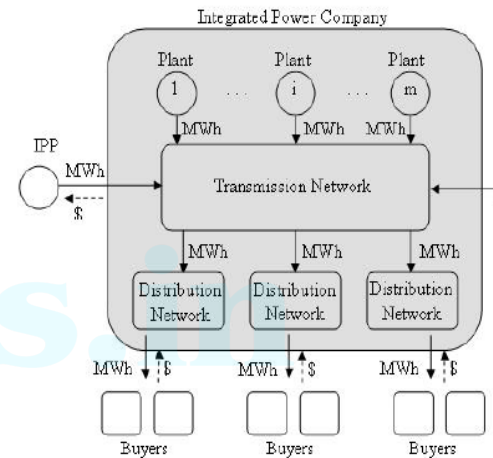
ISSUES IN RESTRUCTURED POWER INDUSTRY

- **Technical issues**
 - ★ Control of transmission grid
 - ★ Scheduling and dispatch services
 - ★ Reactive power supply and voltage control
- **Market related issues**
 - ★ Bidding and trading
 - ★ Type of transaction
 - ★ Level of competition
 - ★ Pricing schemes

Electricity Market Models

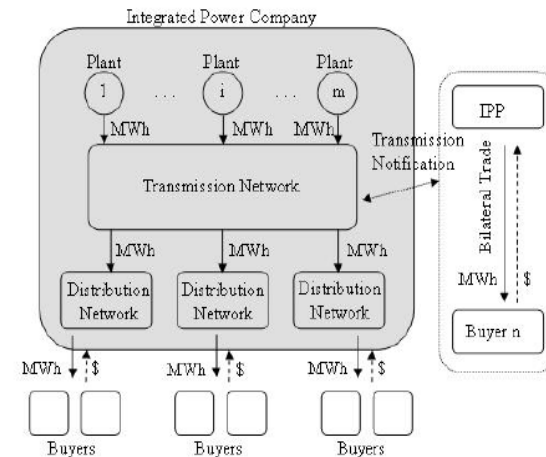
Single-buyer Model

- IPPs compete for electricity supply
- Transmission and distribution remain vertically integrated
- Long-term contracts at fixed prices



Bilateral Contracts Model

- Specific contracts between supplier and consumer
- Appropriate access and transmission pricing standards
- Transfer the traded power over transmission utility



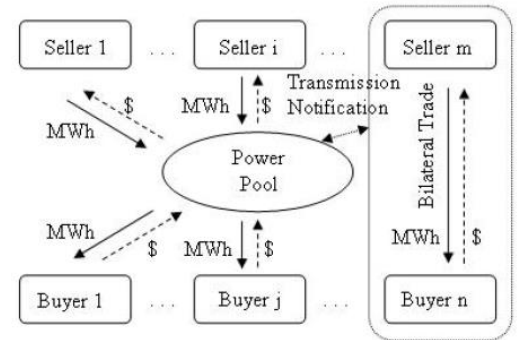
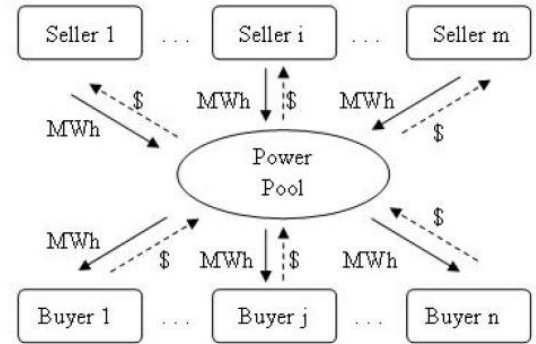
Electricity Market Models

PoolCo Model

- Pool of producers and customers served by transmission system
- Competition among utilities to trade power
- Centralized market place (Px) clears the market and define the Market Clearing Price (MCP)

Hybrid Model

- Combines the features of previous two market models
- Choice of bilateral contracts and power pool
- Power pool serve all buyers and sellers who choose to compete



Distribution Automation (DA)

It is an integrated system concept for the digital automation of distribution sub-station, feeder and user functions. It includes control, monitoring and protection of the distribution system, load management and remote metering of consumer loads. The distribution automation contains:

- Computer hardware
- Computer software
- Remote terminal units (RTUs)
- Communication systems
- Consumer metering devices, relays.

The benefits of DA are:

- Improved quality of supply
- Improved continuity of supply
- Voltage level stability
- Reduced system losses
- Reduced investment

The distribution automation system provides automatic reclosing of relays, automatic feeder switching and provides remote monitoring and controlling of distribution equipment (transformers, capacitors, breakers, sectionalizers, communication nodes etc.) from sub-station up to and including the consumer interface. It affords the utility in minimizing outage time and ultimately, better consumer service and lowering of the total delivered cost of electricity. It allows operation of the system with less capacity margin.

Previously Communication protocol IEC 60870-5, 101/104 and 870-5-101/104 were mostly used for RTU based automation in India. A new communication protocol standard IEC 61850 for data communication between metering, protection, control, transformer and switching devices.

IEC 61850:2003 Communication Networks and Systems in Substations(SCADA)

The Central Electricity Regulatory Commission has emphasized the distribution companies/state electricity boards in India for having their substations to be compatible with SCADA systems. The various functions can be:

- Electrical network analysis
- Work management
- Trouble call analysis
- Consumer load monitoring
- Intelligent remote metering, e.g., automatic meter reading etc.
- Automated capacitor control
- Sub-station automation
- Intelligent electric devices
- Advanced remote terminal units
- Computerized power distribution relays
- Power quality monitoring
- Energy Management

As work forces are being cut, and consumer expectations and demands are rising, automation becomes essential to system maintenance and improvement of consumer service. Some standards for various distribution automation functions such as data formats, communication protocols, etc., are now beginning to emerge.

Remote Terminal Units (RTUs): Modern RTUs are microprocessor based devices and are designed to acquire data and transfer the same to the Master Station through a communication link radio, power line carrier, wire, fibre optic etc. RTUs collect data from transducers, transmitters, contact inputs from equipment/ instruments, meter readings etc., perform analogue/digital conversions, check data-scaling and corrections and send/receive messages from/to master station(s) via interfaces. The RTU is usually designed to monitor parameters such as: Bus-line volts, current, active power, reactive power, status of circuit breakers, switches and isolators, fault detection, temperature, level, pressure, flow, etc.

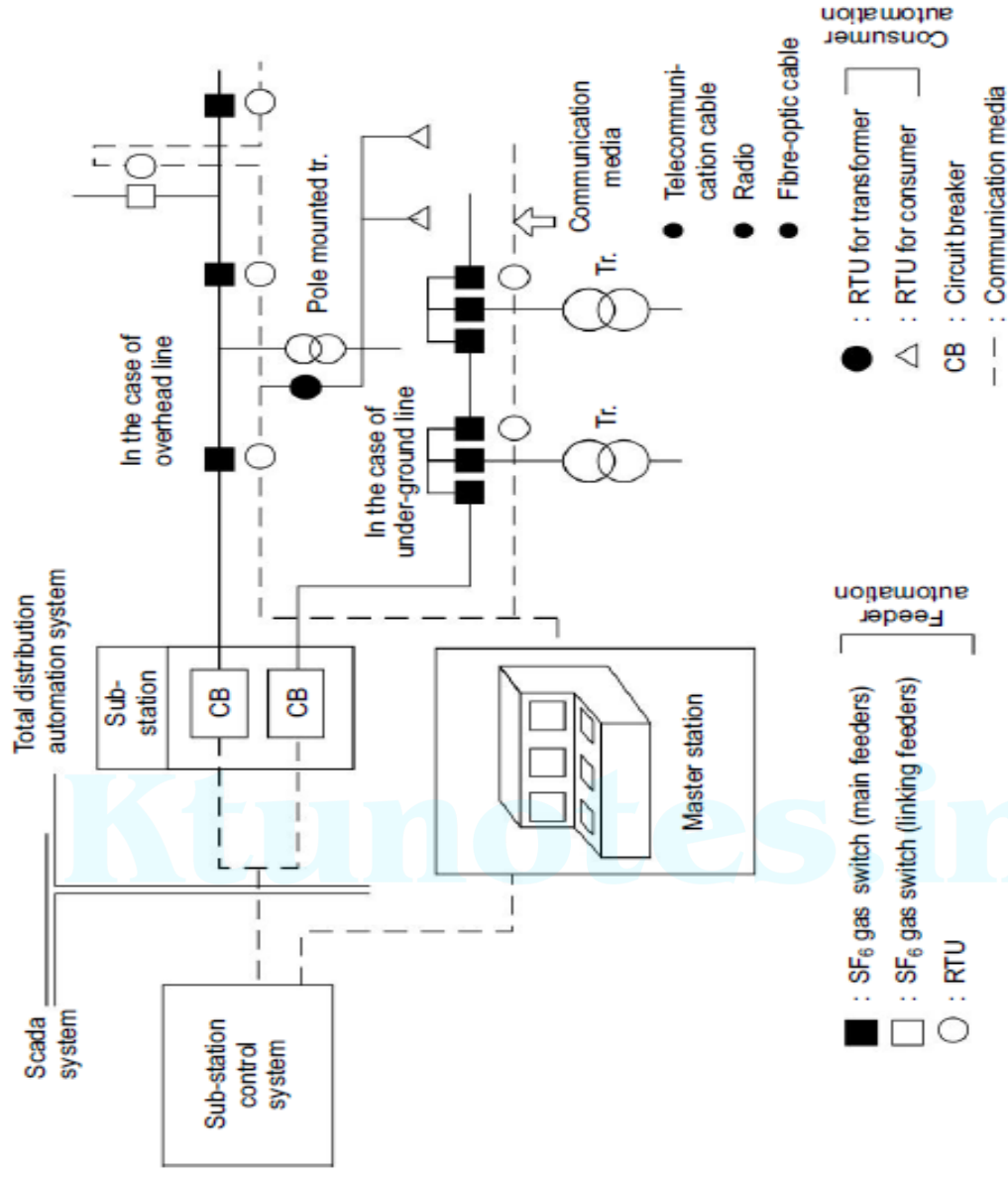


Fig. 5.14 Typical distribution automation system